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Snow loss modeling for solar modules using image processing and deep learning

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ABSTRACT

Energy loss resulting from partial or full snow coverage on solar modules, such as photovoltaics (PV), poses serious challenges to the efficiency of renewable sources in cold climates. This study introduces a new method to quantify the impact of snow on installed PV panels using image processing and deep learning (DL) techniques. A set of 44 PV panels in four different arrays were examined in the experiment through per-pixel analysis that was correlated with various performance indicators. Results of two different PV selection methods, Direct Selection (DS) and Perspective Transformation (PT), were compared, with an acceptable deviation that ranges from 0.1% to 7.35% of one panel area. Energy production was computed where the estimated lowest average was 0.0368 kWh/panel with DS and 0.0395 kWh/panel with PT, due to high uncovered PV ratio with an average value of 84.1%. The validation with alternative methods resulted with an average deviation of 1.21%. Neural networks proved to offer strong correlation between input/output pairs with an average *R* value of 0.9. Additional uncertainty analysis showed that the error for area coverage detection increased due to higher snow deposition rates from 1.8% to 3.7%. Overall, remote monitoring of PV panels can be predicted to enhance energy production performance, while using DL-aided digital photography to track accumulation.

1. Introduction

The rapid urbanization and developments in industry have led to many challenges in energy supply and environmental performance in recent decades [1,2]. Fossil resources such as coal, oil, or natural gas are increasingly depleted and cannot meet the tremendous increase in energy demand [3]. Moreover, massive consumption of carbon-based fuels has released excessive amounts of greenhouse gases, leading to environmental pollution and global climate change. Consequently, the potential of renewable energy sources was extensively investigated [3–7]. In such a context, new materials, advanced modeling and cutting-edge technologies are continuously needed to improve energy efficiency [1].

In 2020, the capacity additions (capacity expansion) of renewable energy increased by 45%, which is the highest ever since 1999. Subsequently in 2021 and 2022, renewable energy was expected to take up to 90% of the global capacity additions of new power [8]. Solar energy stands out among all the renewable resources for its high maturity and wide acceptance [9]. Besides, it far outpaces other sources in growth rate due to cost reduction and scalability [10,11]. Beyond the photovoltaic (PV) farms, building integrated PV has become a promising technology to provide electricity to utility grids while saving land space and preserving the environment [12,13].

In cold climates, the largest energy demand in buildings evolves during the winter due to space heating requirements. However, PV systems in such contexts are subject to snow cover and in some cases to less solar radiation due to high latitudes. This results in snow accumulation which reduces the amount of irradiance that reaches the solar cells, resulting in significant energy loss. The highest annual loss caused by snow coverage has been reported to be 34% of the total generation [14]. During winter months, this loss reaches 90% to 100% of expected generation from PV [15,16]. Typical annual loss can be less than 10% [17]. In [16], the impact of snow deposition on energy loss from PV was 25.8% on average. Such a study performed monthly measurements of the energy loss for 18 months for 4 different tilt angles of PV panels. The data represented 72 samples that can be used to assess the feasibility of applying deep learning in energy loss prediction by snow coverage. Therefore, instead of maximizing overall performance of PV systems, improving the generation capacity and dependability in winter would result in valuable energy contributions to electricity grids during this season.

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Fig. 1. Sample of examined PV conditions collected from different sources.

There are several key studies on the effects of snow coverage on PV systems [18–20]. In [21], satellite imaging was utilized to estimate snow accumulation, while two major errors were found in this method including: (1) False identification where images show uncovered PV and the panels were covered, and (2) Over-estimation of times of snow cover. These errors were 26% and 23% respectively, which provided the probability of errors in prediction. Recently, image processing techniques have been highly utilized to improve the performance of renewable energy applications through analysis, prediction, and optimization [22]. Such techniques aimed to extract useful information by performing a variety of operations on digital images [23,24]. Image processing surpassed other methods because of its simplicity, rapidity, accuracy, nondestructive nature, and feasible cost. For instance, defect detection in PV was conducted by [25,26] using image processing to assess the destructive effects and enhance the reliability of the system. Moreover, shadow recognition approaches based on image processing were proposed by [22,27–29] to facilitate the reconfiguration process of PV arrays and thus increase the energy production by PV systems.

Furthermore, deep learning (DL) techniques have been proved to be powerful in predicting accurately by analyzing a large dataset and extracting the potential interrelationships in the data. Comprehensive models can be designed and trained to be applied to similar tasks after generalization or modification. DL has been widely used in the field of PV performance prediction. For example, PV power generation in various real conditions was forecasted using DL to investigate the short-term performance of PV systems [30,31]. In [32], Levenberg–Marquardt (LM) learning algorithm was found to provide the most accurate 1 day-ahead power production predictions with less computational efforts compared with 9 different learning algorithms. DL has also been used to predict power loss of solar PV due to soiling in both stable and complicated environments, which focused not only on snow coverage but also on accumulation of dust, leaves, and so on [33,34]. Nonetheless, very limited research has considered combining image processing and deep learning techniques together to get enhanced performance.

DL approaches enhanced the forecasting accuracy in solar applications due to their learning potential of time series patterns [35]. However, current approaches had noticeable drawbacks such as using human-crafted or pixel-based features [36–38]. This was tackled by DL encoding techniques that employed classification models for feature extraction such as clear sky index classifiers [39]. Other DL algorithms examined mutual interactions between climatic conditions and achieved an improved accuracy of 18.34% compared to artificial neural networks [40]. Convolutional Neural Networks (CNNs) were also efficient in image processing due to their learning capabilities and effective weight-sharing procedures [41]. CNN algorithms had an effective performance in relation to extracting intrinsic and nonlinear features by convolutional layers with the appropriate pooling operations [42]. CNNs were also employed for real energy forecasting and overcame performance deficiencies related to power reliability in real-time monitoring systems [43]. However, very limited research has considered applying CNNs in solar forecasting due to the limited feature dimension of CNNs which restricted their potential as local feature extractors [43].

In order to quantify the energy loss due to snow coverage, it was critical to evaluate the effective uncovered area of the PV panel, and then adjust the PV system with electrical circuit reconfiguration strategies to achieve maximum power generation efficiency [22,44]. Various snow loss models were proposed to predict the impact of snow coverage. For example, a daily snow loss model was developed in [45] based on a correlation between daily snow losses and factors including tilt angle, temperature, irradiance, and snow depth. Moreover, in [16], a monthly snow loss model was proposed based on a correlation between monthly snow losses and parameters such as snowfall quantity, tilt angle, ground interference, and climate (wind, temperature, humidity, radiation, etc.). However, these models were built on the basis of empirical correlations between snow losses and ambient conditions [46]. They were highly dependent on specific locations and climates, and cannot be applied universally to all situations with high accuracy. Consequently, snow detection was more commonly used than snow loss models in PV monitoring [46].

Using image processing and recognition techniques, this paper develops a new method to detect snow coverage on PV panels in order to quantify the impact of accumulation on energy loss. While determining the probability distribution of deposition on the panels, the intent is to identify factors that influence the generation deficiency using real-time images to analyze the PV system. Through this approach, the paper seeks to find an accurate and effective method for identifying snow deposition and calculating uncovered rates of PV areas to remotely monitor and model the energy production.

2. Methodology

This section presents the method used to examine the impact of snow coverage on PV panels. The image data was first collected and classified into different conditions of PV with a complex background of building elements, trees, fields, sky, snow, shadow, and orientation. Six conditions of snow coverage were studied, including: PV and snow; PV on buildings with sky and snow; PV on buildings with ground and snow; PV on nearby buildings with sky, ground and snow; PV on distant buildings with sky, ground and snow; and PV farm with sky, ground and snow. The size of the images in the dataset varied between a small size of 474 × 308 pixels and a large size of 4256 × 2832 pixels. Fig. 1 provides a representative sample of examined PV conditions.

The PV conditions were then analyzed based on a three-step computational algorithm, including: (1) Pre-processing of image-data, (2) PV panel selection procedure, and (3) Actual PV area calculation (Fig. 2). Deep learning was finally utilized to verify the PV performance prediction using the Levenberg–Marquardt Backpropagation (LMB) and Convolutional Neural Network (CNN) methods for neural network training. Both image processing and

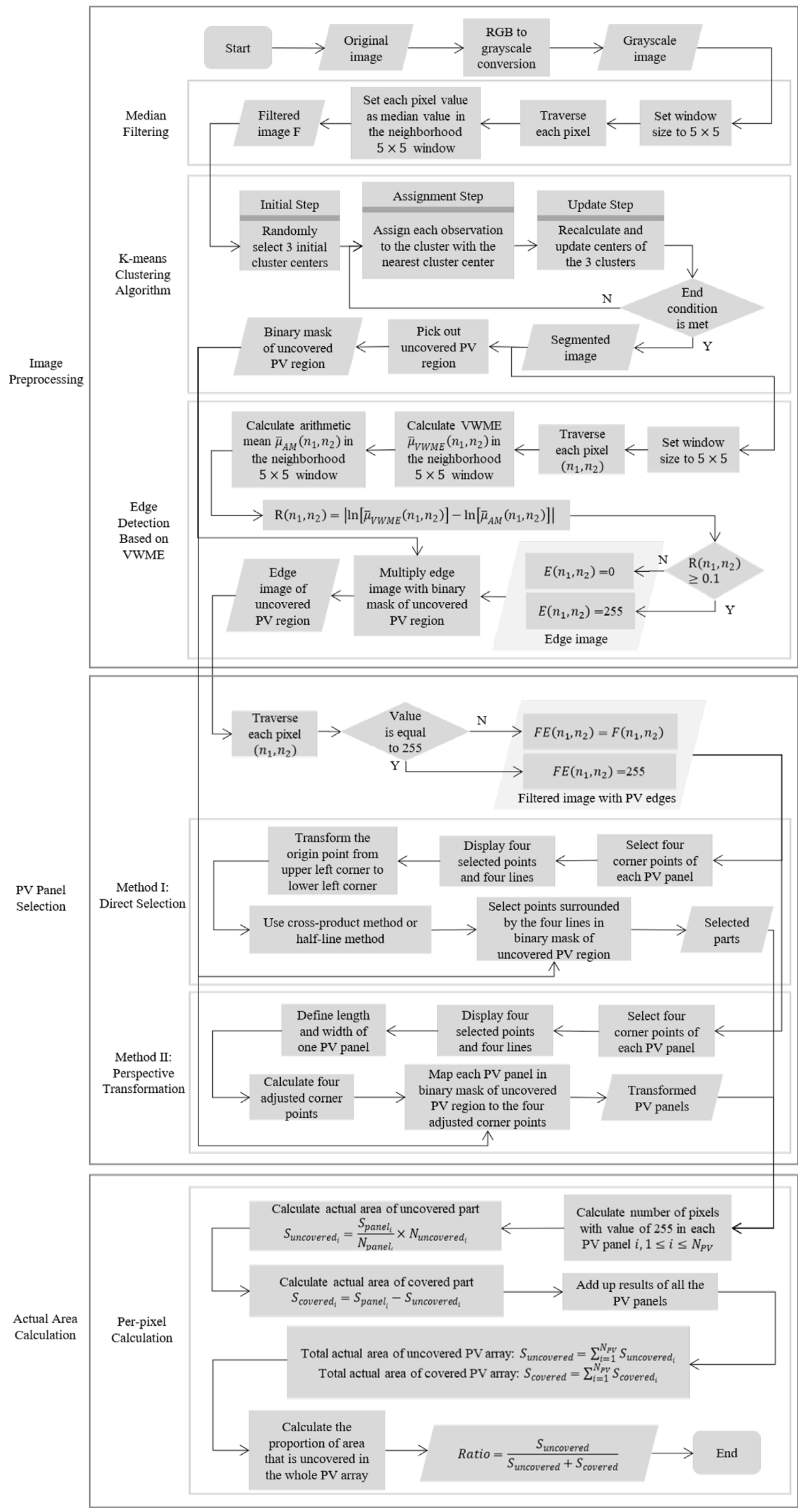


Fig. 2. Flow chart of the method.

96	94	92	93	96	91	98
95	96	93	97	92	99	91
82	95	84	90	89	96	93
97	85	89	92	96	92	89
95	94	98	96	93	91	91
95	93	92	94	95	93	89
88	90	95	93	92	96	91

Fig. 3. Example of median filtering with window of 5 × 5. The median of neighborhood values is 93.

deep learning were realized using MATLAB toolbox. Through this procedure, a set of 44 PV panels in four different PV arrays were examined through per-pixel analysis that was correlated with performance indicators, including: series resistance, image masking techniques, and PV type and efficiency which compared the PV generation to the global irradiation incident on the panel.

2.1. Pre-processing of image-data

2.1.1. Eliminating isolated noise

The processing algorithm started by converting the RGB images into grayscale ones. These images were taken from the ground and usually contained unnecessary or redundant interference information, which seriously affected the quality of them. Therefore, median filtering was performed with a window of 5 × 5 to overcome any random noise in the images. Such nonlinear smoothing technique set the gray value of each pixel as the median of all pixels in the neighborhood window according to the mathematical model:

$$g\left(x, y\right) = med\{f\left(x-k, y-l\right), (k, l)\epsilon W\} \tag{1}$$

where $f(x, y)$ and $g(x, y)$ are the original image and filtered image, respectively. W is the 5 × 5 window (Fig. 3). Consequently, the gray value of each pixel was close to its surrounding points, thus eliminating isolated noise points while protecting the edges of the image from being blurred.

2.1.2. Iterative refinement through K-means clustering

To further extract the objects of interest, the images were divided into regions with different characteristics using a K-means clustering algorithm. The algorithm maximized intra-cluster similarity and minimized inter-cluster similarity, as an iterative refinement technique. For n observations in data set D, the center of each cluster was represented by the mean of all observations in the cluster, through:

(1) Initial step: K observations were randomly selected as the initial cluster centers, $m_1^{(1)}, \ldots, m_K^{(1)}$ in D.

(2) Assignment step: The squared Euclidean distance between each observation and each selected cluster center was determined. Afterward, each observation was assigned to exactly one cluster with the nearest cluster center. For $1 \leq p \leq n$, $1 \leq i, j \leq K$,

$$S_i^{(t)} = \left\{x_p : \left\|x_p - m_i^{(t)}\right\|^2 \leq \left\|x_p - m_j^{(t)}\right\|^2\right\} \tag{2}$$

where x_p is the pth observation. $S_i^{(t)}$ is the ith cluster.

(3) Update step: Observations were assigned to a cluster comprising cluster centers. Therefore, the means of the clusters were recalculated according to the existing observations in the clusters.

$$m_i^{(t+1)} = \frac{1}{\left|S_i^{(t)}\right|} \sum_{x_j \in S_i^{(t)}} x_j \tag{3}$$

where $m_i^{(t+1)}$ is the recalculated ith cluster center.
(4) Steps 2 and 3 were performed alternately until the termination condition was met. Consequently, the clusters formed in this round were the same as those formed in the previous one.

The entire filtered image formed a data set by viewing the gray value of each pixel in the image as an observation. Then K-means clustering algorithm led to image segmentation (Fig. 4). The image included three regions: (1) Uncovered PV region (darkest), (2) Covered PV region (lightest), and (3) Background region: As a result, K value was set to three and the image was segmented into three color clusters. For the purpose of background deletion, the binary mask (element 1 for uncovered PV region and 0 for the remaining regions) was created from the segmented image. By multiplying this mask with the original image, the uncovered PV region could be obtained.

2.1.3. Edge detection of PV panel

In an image, an edge was identified as the abrupt transition between regions of different intensities. To assist the subsequent selection of PV panels, the edges of the PV array were detected from the segmented image. The edge detection method was based on Variance-Weighted Mean Estimate (VWME) [47], applied as follows:

(1) An $n \times n$ square window centered at pixel (n_1, n_2) was defined, where the pixels inside the window were divided into eight groups:

$$\{x\left(n_1 - i, n_2 - j\right) \mid (i, j) \in s_p, p = 1, 2, \ldots, 8\} \tag{4}$$

where s_p are the eight subsets as shown in Fig. 5.

(2) The mean and the corresponding variance of the image intensity in each of the eight subsets were calculated.

(3) The variance-weighted mean estimate was calculated according to:

$$\bar{\mu}_{VWME}\left(n_1, n_2\right) = \frac{\sum_{p=1}^{8} b_p\left(n_1, n_2\right)\bar{\mu}_p\left(n_1, n_2\right)}{\sum_{p=1}^{8} b_p\left(n_1, n_2\right)},$$
$$b_p\left(n_1, n_2\right) = \frac{1}{\bar{\sigma}_p^2\left(n_1, n_2\right)} \tag{5}$$

where $\bar{\mu}_p\left(n_1, n_2\right)$ and $\bar{\sigma}_p^2\left(n_1, n_2\right)$ are the mean and variance of the image intensity in the pth subset.

(4) The arithmetic mean of the image intensity in the entire $n \times n$ square window was calculated and denoted as $\bar{\mu}_{AM}\left(n_1, n_2\right)$.

(5) The variance-weighted mean estimate $\bar{\mu}_{VWME}\left(n_1, n_2\right)$ was compared with the arithmetic mean $\bar{\mu}_{AM}\left(n_1, n_2\right)$. For the central pixel (n_1, n_2),

$$R\left(n_1, n_2\right) = \left|\ln\left[\bar{\mu}_{VWME}\left(n_1, n_2\right)\right] - \ln\left[\bar{\mu}_{AM}\left(n_1, n_2\right)\right]\right| \tag{6}$$

where $R\left(n_1, n_2\right)$ is the decision equation. If the window was in a uniform region, $\bar{\mu}_{VWME}\left(n_1, n_2\right)$ was similar to $\bar{\mu}_{AM}\left(n_1, n_2\right)$. As a result, $R\left(n_1, n_2\right)$ was close to zero. However, if an edge was there, the abrupt change in the variance led to relatively higher $R\left(n_1, n_2\right)$.

(6) The threshold of $R\left(n_1, n_2\right)$ was set to 0.1. If $R\left(n_1, n_2\right) \geq 0.1$, the edge image element at (n_1, n_2) was set to one. Otherwise, the edge image element at (n_1, n_2) was set to zero.

(7) Steps 2 to 6 were repeated for another pixel (n_1, n_2) until the whole image was covered (Fig. 5).

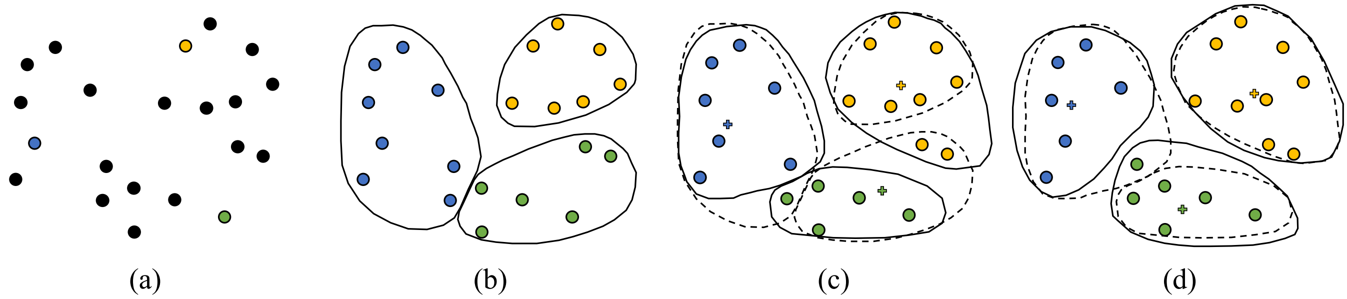


Fig. 4. Process of K-means clustering algorithm in the case of $K = 3$. (a) Step-1, (b) Step-2, (c) Step-3, and (d) Step-4. . (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

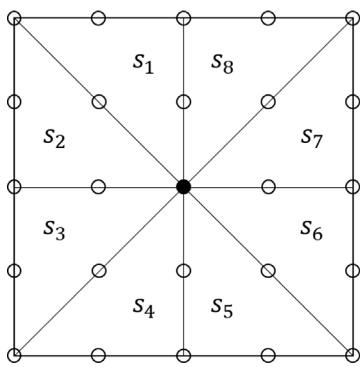


Fig. 5. Eight subsets of the $n \times n$ square window.

When all the pixels in one of the eight subsets were of equal image intensity, the mean of the image intensity in that subset was equal to the intensity of each pixel, and the corresponding variance was zero. Consequently, the variance reciprocal approached infinity. In this case, the equation of VWME was interpreted in the limit sense.

Given m subsets ($m \leq 8$) of the eight subsets having pixels with the same image intensity as that of the central pixel, each subset i ($1 \leq i \leq m$) had an image intensity mean, variance, and variance reciprocal of $mean_i(n_1, n_2)$, $var_i(n_1, n_2)$, and $b_i'(n_1, n_2)$, respectively. For the remaining $(8 - m)$ subsets, the variance was higher than zero. Therefore, the reciprocal of each variance was a finite number. Furthermore, the product of each variance reciprocal and its corresponding mean was also a finite number. The variance-weighted mean estimate was then calculated from:

$$\bar{\mu}_{VWME}(n_1, n_2) = \frac{C_2 + \sum_{i=1}^{m} b_i'(n_1, n_2)\, mean_i(n_1, n_2)}{C_1 + \sum_{i=1}^{m} b_i'(n_1, n_2)}, \tag{7}$$

$$b_i'(n_1, n_2) = \frac{1}{var_i(n_1, n_2)} \tag{8}$$

where C_1 is the sum of the reciprocals of the $(8 - m)$ variances. C_2 is the sum of the products of the $(8 - m)$ reciprocals of variances and their corresponding mean values. In each of the m subsets, the mean of the image intensity was equal to the intensity of the central pixel. Moreover, the corresponding variance of the image intensity was zero. As a result, the reciprocals of the m variances were equal and approached infinity. The simplified formula was expressed as:

$$\bar{\mu}_{VWME}(n_1, n_2) = \frac{\frac{C_2}{b_1'(n_1,n_2)} + \sum_{i=1}^{m} mean_i(n_1, n_2)}{\frac{C_1}{b_1'(n_1,n_2)} + m} \tag{9}$$

where the same reciprocals of variances were substituted by $b_1'(n_1, n_2)$. As $b_1'(n_1, n_2)$ approached infinity, both $\frac{C_2}{b_1'(n_1,n_2)}$ and $\frac{C_1}{b_1'(n_1,n_2)}$ were zeros, and the final form of the equation was:

$$\bar{\mu}_{VWME}(n_1, n_2) = \frac{\sum_{i=1}^{m} mean_i(n_1, n_2)}{m} = intensity\ of\ (n_1, n_2) \tag{10}$$

where (n_1, n_2) is the central pixel of the $n \times n$ square window. In conclusion, if there were more than one subset of the eight subsets with all pixels having the same intensity, the VWME at the central pixel was equal to the image intensity value of that pixel. After edge detection, the edges of the uncovered PV region were picked out by multiplying the edges of the whole image by the binary mask of the uncovered PV region generated by the K-means clustering algorithm.

2.2. PV panel selection procedure

The binary mask generated by the K-means clustering algorithm was used in the calculation step. To calculate the actual area of both uncovered and covered PV regions, each PV panel was selected from the image. Specifically, the median filtered image was combined with the edge image of the uncovered PV region where the four corner points of each PV panel were selected. Consequently, the four edge lines were formed by connecting these four points. The study proposed two ways to isolate the PV panel: (1) Direct selection, or (2) Perspective transformation, which was an optimization of the former.

2.2.1. Direct selection

Direct selection can be applied using the following two methods:

(1) Cross-product method: To determine whether a point is inside a convex quadrilateral or not, cross-product was used with four sides of the quadrilateral in a clockwise or counterclockwise direction. If point P is inside the quadrilateral, then $\overrightarrow{AP}$, $\overrightarrow{BP}$, $\overrightarrow{CP}$ and $\overrightarrow{DP}$ will be in the clockwise direction of $\overrightarrow{AB}$, $\overrightarrow{BC}$, $\overrightarrow{CD}$ and $\overrightarrow{DA}$ respectively. As a result, the direction of $\overrightarrow{AB} \times \overrightarrow{AP}$, $\overrightarrow{BC} \times \overrightarrow{BP}$, $\overrightarrow{CD} \times \overrightarrow{CP}$ and $\overrightarrow{DA} \times \overrightarrow{DP}$ will be the same. In Fig. 6(a), the four cross-products of vectors point inward perpendicular to the plane. If point P is on one of the four sides, the magnitude of one cross-product is zero, while the others will remain in the same direction. In Fig. 6(b):

$$\left|\overrightarrow{AB} \times \overrightarrow{AP}\right| = \left|\overrightarrow{AB}\right| \left|\overrightarrow{AP}\right| sin\left\langle \overrightarrow{AB}, \overrightarrow{AP} \right\rangle = 0 \tag{11}$$

where $\overrightarrow{AB}$ and $\overrightarrow{AP}$ have the same direction. Especially, if point P coincides with point A, B, C or D, the magnitude of two cross-products will become zero, while the others will remain in the

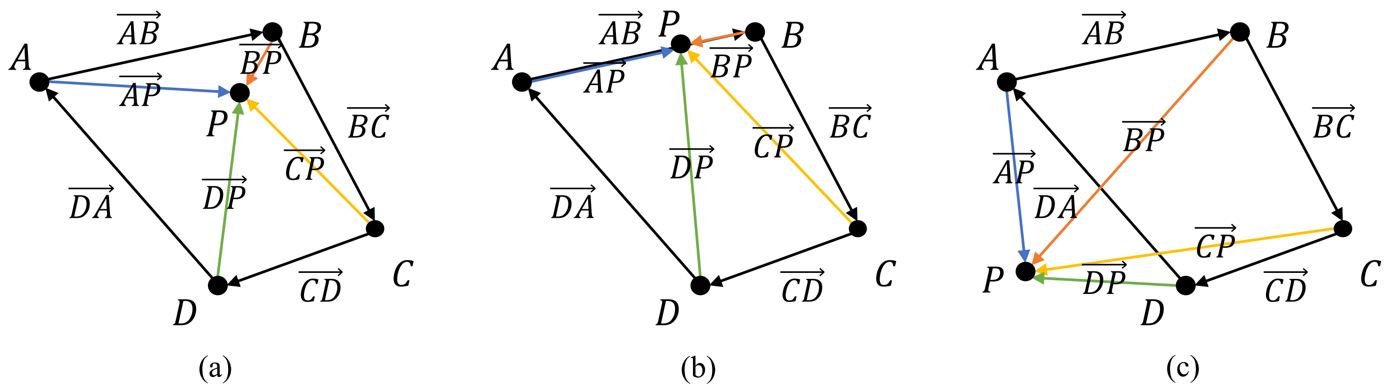


Fig. 6. Schematic diagram of the cross-product and cross-product method. (a) Point P inside the quadrilateral. (b) Point P on the quadrilateral. (c) Point P outside the quadrilateral.

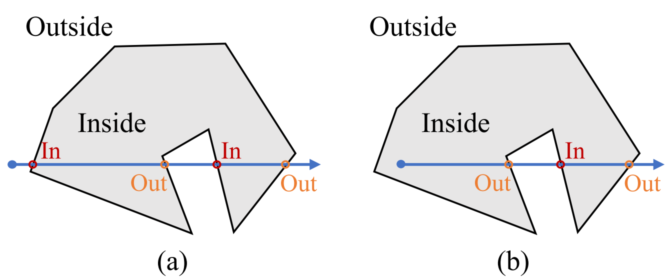


Fig. 7. Schematic diagram of half-line method. (a) An even number of intersection points. (b) An odd number of intersection points.

same direction. If point P is outside the quadrilateral, the direction of $\overrightarrow{AB} \times \overrightarrow{AP}$, $\overrightarrow{BC} \times \overrightarrow{BP}$, $\overrightarrow{CD} \times \overrightarrow{CP}$ and $\overrightarrow{DA} \times \overrightarrow{DP}$ are not the same. In Fig. 6(c), $\overrightarrow{AB} \times \overrightarrow{AP}$, $\overrightarrow{BC} \times \overrightarrow{BP}$ and $\overrightarrow{CD} \times \overrightarrow{CP}$ point inward perpendicular to the paper, while $\overrightarrow{DA} \times \overrightarrow{DP}$ points outward perpendicular to the plane.

(2) Half-line method: For a polygon in the Euclidean plane, the plane was divided into an interior region and an exterior region. A half-line was drawn horizontally from the considered point. Afterward, the number of intersection points between the half-line and each side of the polygon was determined. If the number of intersection points was even, the point was located outside the polygon as shown in Fig. 7(a). Otherwise, the point is located inside the polygon as shown in Fig. 7(b).

Given that all the sides are line segments, the challenging part of the half-line method was to determine accurately whether the half-line intersected each side of the polygon. Therefore, the method was simplified to determine whether a half-line intersects a line segment. Several positional relationships between half-lines and line segments were stipulated to be excluded. Table 1 shows all the considered cases to determine whether a half-line intersects a line segment.

For a point located inside or outside the polygon, the number of intersection points was calculated after determining whether the half-line intersected each side of the polygon. Finally, the point location was determined to be either inside or outside the polygon according to the number of intersection points.

After obtaining the coordinates of the four corner points by selecting in clockwise or counterclockwise direction, and applying the cross-product method or the half-line method, each pixel was categorized to be inside, outside, or on the edge lines of the selected PV panel. Finally, the pixels inside and on the edge lines were picked out from the binary mask of the uncovered PV region.

2.2.1.1. Transformation of coordinates. Selection of the points yielded their coordinates with respect to the origin shown in Fig. 8(a). However, plane geometry calculations used coordinates with respect to the origin shown in Fig. 8(b). As a result, the coordinates were transformed using the equations:

$$x' = x, \;\; y' = H - y + 1 \tag{12}$$

where H is the height of the image.

2.2.2. Selection with perspective transformation

Direct selection of points isolated each PV panel from the original image. However, it did not consider the perspective effects and their associated error. To use the number of pixels for area calculation, the actual area per pixel should be a constant value. Such a fact necessitated the application of perspective transformation of isolated PV panels.

Perspective transformation, or projective mapping, is the projection of an image onto a new visual plane (Fig. 9). By selecting the four corner points of each PV panel in the order: upper left, upper right, lower left, and lower right, the coordinates of the four points were obtained. All the PV panels had almost the same size. Therefore, according to the defined length and width of each PV panel, the coordinates of the four adjusted corner points were calculated. The PV panels with perspective effects were projected onto a new visual plane as standard rectangles with the correct proportion. At this point, the actual area per pixel was the same, and the subsequent actual area calculation was accurate.

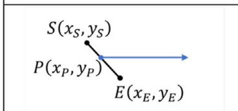
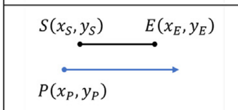
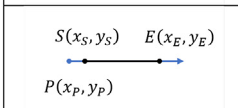
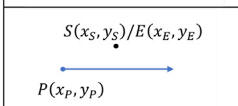
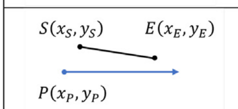
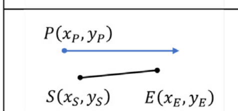
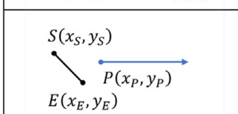
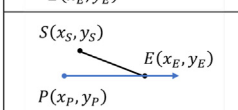
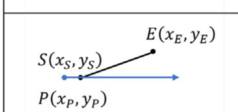
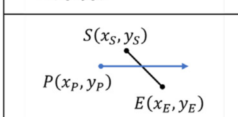
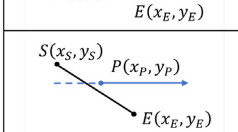
2.3. Actual PV area calculation

The area of examined PV panels was 1.6 m^2. For each selected PV panel, the uncovered PV area was calculated from:

$$S_{uncovered_i} = \frac{S_{panel_i}}{N_{panel_i}} \times N_{uncovered_i}, \;\; 1 \le i \le N_{PV} \tag{13}$$

where S_{panel_i} is the PV panel area, m^2. $S_{uncovered_i}$ is the uncovered PV area, m^2. N_{panel_i} is the pixels' number of the PV panel. $N_{uncovered_i}$ is the pixels' number of the uncovered PV part. N_{PV} is the number of PV panels included in the image. The actual area of the covered

Table 1
Considered cases to determine whether a half-line intersects a line segment. P is the point to be determined, S and E are the start and endpoints of the line segment.

Case	Description	Judgment Condition	Treatment
	P is on the line segment.	$\frac{y_E-y_P}{x_E-x_P}=\frac{y_P-y_S}{x_P-x_S}$	P is on the sides.
	Special case: P coincides with S or E.	Compare the coordinates.	
	The half-line is parallel to the line segment.	$y_S=y_E$	Disjoint and excluded.
	The half-line and the line segment overlap.		
	The length of the line segment is zero.		
	The half-line is below the line segment.	$\begin{cases} y_S>y_P \\ y_E>y_P \end{cases}$	
	The half-line is above the line segment.	$\begin{cases} y_S<y_P \\ y_E<y_P \end{cases}$	
	The half-line is to the right of the line segment.	$\begin{cases} x_S<x_P \\ x_E<x_P \end{cases}$	
	The intersection point is the lower end of the line segment $E(x_E,y_E)$.	$\begin{cases} y_S>y_P \\ y_E=y_P \end{cases}$	
	The intersection point is the lower end of the line segment $S(x_S,y_S)$.	$\begin{cases} y_S=y_P \\ y_E>y_P \end{cases}$	
	The half-line and the line segment have intersection points.	$x_E-\frac{y_E-y_P}{y_E-y_S}*(x_E-x_S)>x_P$	Included.
	The half-line and the line segment do not have intersection points.	$x_E-\frac{y_E-y_P}{y_E-y_S}*(x_E-x_S)<x_P$	Disjoint and excluded.

PV part was calculated by subtracting the actual area of the uncovered PV part from the whole PV panel:

$$S_{covered_i}=S_{panel_i}-S_{uncovered_i},\quad 1\le i\le N_{PV} \tag{14}$$

where $S_{covered_i}$ is the covered PV area, m^2. By repeating the previous steps for all PV panels, the total areas and their associated ratio were calculated from:

$$TS_{uncovered}=\sum_{i=1}^{N_{PV}}S_{uncovered_i}, \tag{15}$$

$$TS_{covered}=\sum_{i=1}^{N_{PV}}S_{covered_i}, \tag{16}$$

$$R=\frac{TS_{uncovered}}{TS_{uncovered}+TS_{covered}} \tag{17}$$

where $TS_{uncovered}$ is the total actual area of the uncovered PV array, m^2. $TS_{covered}$ is the total actual area of the covered PV array. R is the proportion of area that is uncovered in the whole PV array.

2.4. Deep learning

2.4.1. Levenberg–Marquardt backpropagation

Neural network training can advance the PV performance prediction very accurately, while taking reasonable time to handle large data sets. Through this application, the Levenberg–Marquardt Backpropagation (LMB) method was chosen for the

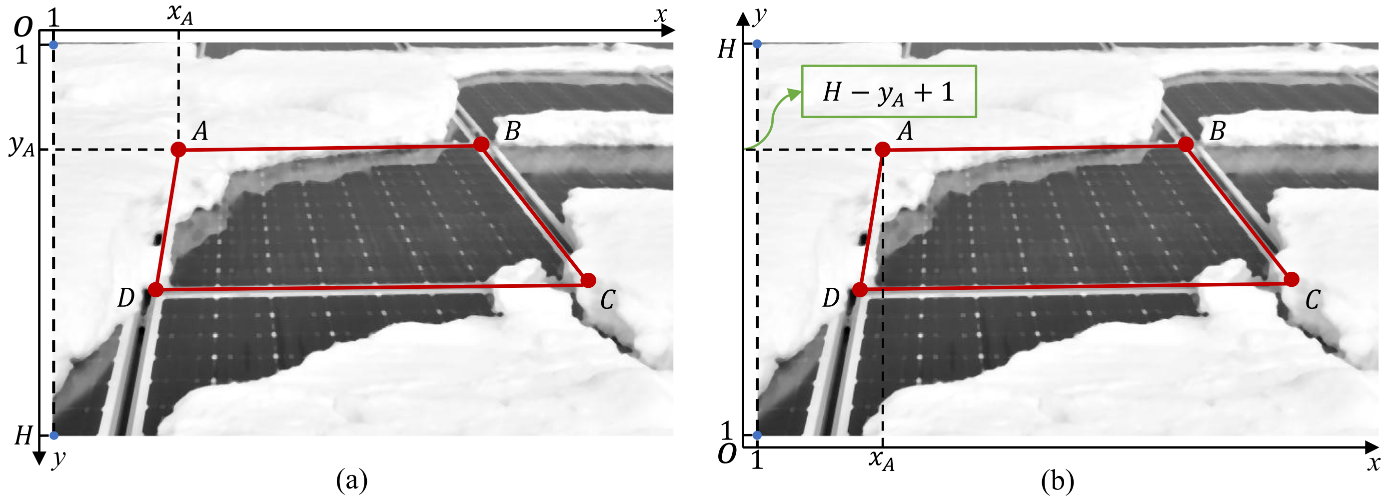


Fig. 8. Transformation of coordinates. (a) Coordinates before transformation. (b) Coordinates after transformation.

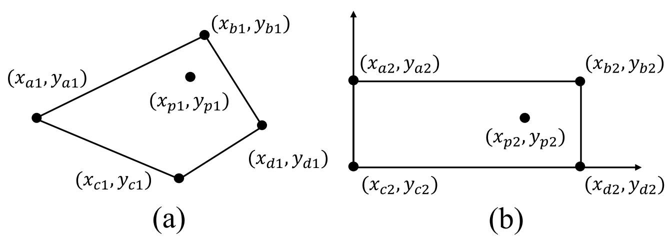


Fig. 9. Perspective transformation. (a) Before transformation. (b) After transformation.

training algorithm setup. This method is widely applied in the problem of least-squares curve fitting. Given m pairs of independent and dependent variables (x_i, y_i), it tries to find parameter β of model curve $f(x, \beta)$ to minimize the sum of the squares of the deviations $S(\beta)$:

$$\hat{\beta} \epsilon \underset{\beta}{argmin} S(\beta) \equiv \underset{\beta}{argmin} \sum_{i=1}^{m} [y_i - f(x_i, \beta)]^2 \tag{18}$$

which is assumed to be non-empty. Such an algorithm usually requires less time at the expense of higher memory.

Data of PV monitoring were imported from [16]. Month and tilt angles in addition to boundary conditions of dry bulb temperature, global horizontal radiation, direct normal radiation, diffuse radiation, wind speed, wind direction, and sky cover were considered in the algorithm training. Overall, the input data comprised 72 samples of 9 elements, while target data comprised 72 samples of 1 element representing the energy loss. Training, validation, and testing were assigned 70% (50 samples), 15% (11 samples), and 15% (11 samples) of the whole sample, respectively. The neural network structure had 3 layers with 10 hidden neurons. Training samples were fed to the algorithm during training, where the network was adjusted according to their error. Validation samples were used to quantify network generalization, and to stop the training when generalization was not improving anymore. Testing samples were used to measure independently the performance after training. Thus, they did not have any impact on training.

2.4.2. Convolutional neural network

A regression model was further fitted using a Convolutional Neural Network (CNN) to compare its performance to LMB algorithms. A CNN architecture was constructed and a network was trained to predict the energy loss due to snow coverage. The data set contained the input array of month and panel tilt together with the corresponding energy loss as the output. Data were normalized in every phase of the network to stabilize and accelerate the training by employing gradient descent. This was necessary to avoid poorly scaled data where the loss became *Not a Number*, i.e., 0/0 or inf/inf. In this case, the parameters of the CNN experienced a significant divergence during the network training. Data normalization was thus required by rescaling where the corresponding range became [0, 1]. Normalization was performed by

$$x_n = \frac{x - x_2}{x_1 - x_2} \tag{19}$$

where x_n is the normalized data point. x_1 and x_2 are the maximum and minimum data points in the trained set. Predictors or input data were normalized to the same range before admission to the CNN. Outputs of each layer, either convolutional or fully connected, were normalized by employing a batch normalization layer. The formula of batch normalization was expressed by

$$z^N = \left(\frac{z - m_z}{s_z}\right) \tag{20}$$

where z^N and z are the normalized and regular outputs of the neurons. m_z and s_z are the mean and standard deviation of the neurons' outputs. Fig. 10 shows a schematic of the employed CNN. x_i, z, a, and y represent inputs, neurons' outputs, activation functions' outputs, and network output.

Batch norm represented with the tilted gray line was applied to the neurons' outputs before employing the activation function. Neurons without batch normalization were calculated from

$$z = g(\omega, x) + b, \ \ a = f(z) \tag{21}$$

where g is the neuron's linear transformation. ω is the neuron's weight. b is the neuron's bias. f is the activation function. Both ω and b were updated during the training process. Neurons with batch normalization were calculated from

$$z = g(\omega, x), \ \ z^N = \left(\frac{z - m_z}{s_z}\right) \cdot \gamma + \beta, \ \ a = f\left(z^N\right) \tag{22}$$

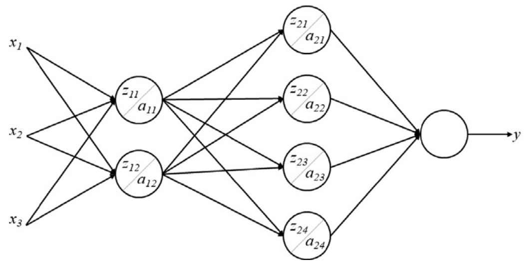


Fig. 10. CNN employed for energy loss prediction due to snow deposition.

Table 2
Training options for the employed CNN.

Training option	Value
Minimum batch size	128
Maximum epochs	30
Initial learn rate	0.001
Learn rate drop factor	0.1
Learn rate drop period	20

where γ and β are the learning parameters of the batch normalization. This step guaranteed normalization of the predictions of the CNN when the training was initiated. However, the network training diverged when the scale of both response and predictions was noticeably different. In this case, the response was normalized to avoid poor scaling and the CNN training was monitored for potential improvements. Response normalization prior to training necessitated the transformation of the CNN predictions to get them within the same scale as the original response. Although exact normalization of data was not required, training the CNN to predict $100 \times y$ or $y + 500$ instead of y lead to the divergence of the CNN parameters when the training was initiated. Such results were observed even though the only difference between a CNN targeting either $a \times y + b$ or y was a simple rescaling of the network parameters. In the case of skewed or uneven input/response, nonlinear transformations were performed by taking logarithms to the input data prior to training the CNN. The regression problem was solved by creating the network layers and including a regression layer at the end of the CNN. Input data type and size were defined in the first layer. Such inputs were 1-by-2-by-1. Core architecture of the CNN was then defined in middle layers of the network. These layers were responsible for most of the learning/computation taking place. Output data type and size were defined in the final layer. All previous layers were combined together in a given array. MATLAB was employed for training the CNN using the options shown in Table 2.

3. Results and discussion

The main results from this study demonstrated the magnitude of energy generation losses due to snow accumulation on various conditions of PV panels. The impact of different environments, backgrounds, PV array scales, and snow cover forms were investigated. Using the proposed procedure, some inferences about the effect of snow coverage on PV can be derived from this diverse application.

3.1. Characteristics of examined images

Fig. 11 shows the algorithm performance of the six representative images. Considering the three evaluation criteria in the algorithm, C_1 for segmentation that divided the image into uncovered PV region, covered PV region, and background region; C_2 for binary mask of the uncovered PV region generated from the segmented image; and C_3 for isolation that extracted each PV panel from the image, the following can be stated:

In Fig. 11(a) and (b), the three criteria were perfectly satisfied despite the images' complexity. This can be interpreted by PV panels' appropriate sizes, clear boundaries, sharp color contrast, and the PV corner-point inclusion inside the images. In Fig. 11(c), the image was properly segmented into three regions. However, there were some problems with the generation of the binary mask. The uncovered PV region normally had the darkest color in the image. Consequently, the darkest region was picked out from the segmented image when generating the binary mask of the uncovered PV region. Both the uncovered and covered PV regions were close in color in such a case. Besides, the color of the house was the darkest among all the elements. Therefore, the algorithm picked out the wrong region and created an inaccurate binary mask that led to a wrong estimate of the actual area finally.

The algorithm performed effectively for both Fig. 11(e) and (f) regarding segmentation and binary mask generation. However, it failed to isolate the PV panels from the image. In Fig. 11(e), the challenges that hindered the PV isolation were: (1) The small area occupied by the PV array, (2) The large portion occupied by the sky, snow, and the house, and (3) the blurred boundaries between PV panels. In Fig. 11(f), the isolation step was unsuccessful due to the extrusion of PV panels from the image where some corner points were outside the image borders. Fig. 11(d) had the worst results among the investigated set. The current algorithm failed to perform any of the three steps successfully. That was mainly because of the differences in the color of the uncovered PV region resulted from reflections. Consequently, the uncovered PV region was segmented into two colors by the K-means clustering algorithm, which led to inaccurate segmentation. Generation of the binary mask took out only the darker part of the uncovered PV region. Moreover, the fuzzy boundaries restrained the selection step.

Overall, the algorithm performed effectively when the image had the following characteristics: (1) The PV array was centered and occupied most area of the image, (2) Part of the PV array was covered by snow, while the remaining part was bare, (3) The PV array contained multiple PV panels of appropriate sizes, which were separated by frames that preserved clear boundaries between the panels, (4) The color of the uncovered PV region was uniform and dark, (5) There was a considerable color contrast between the PV panels and the surroundings, and (6) The corner points of all PV panels were included inside the image.

3.2. Comparison of selection

Fig. 12 shows the image-processing sequence for the four arrays, while Fig. 13 compares the direct selection output and the perspective transformation output. The former resulted in few distorted panels due to various factors such as the shooting angle of the original images with direct selection. Consequently, the accuracy of this approach was lower than the perspective transformation approach. Overall, the mean absolute error (MAE) and the root mean square error (RMSE) were 2.68% and 3.25% of the PV area, respectively. The deviation between the two approaches ranged from 0.1% to 7.35% of the PV area, in the 9th panel of *Array 2* and the 4th panel of *Array 1*, respectively. Overall, such deviation range was acceptable, and the accuracy of both approaches was relatively high, despite the slight distortion in the direct selection approach.

Fig. 14 shows the energy production loss (E_{loss}) for the examined panels in each array. E_{loss} was represented by kWh based on

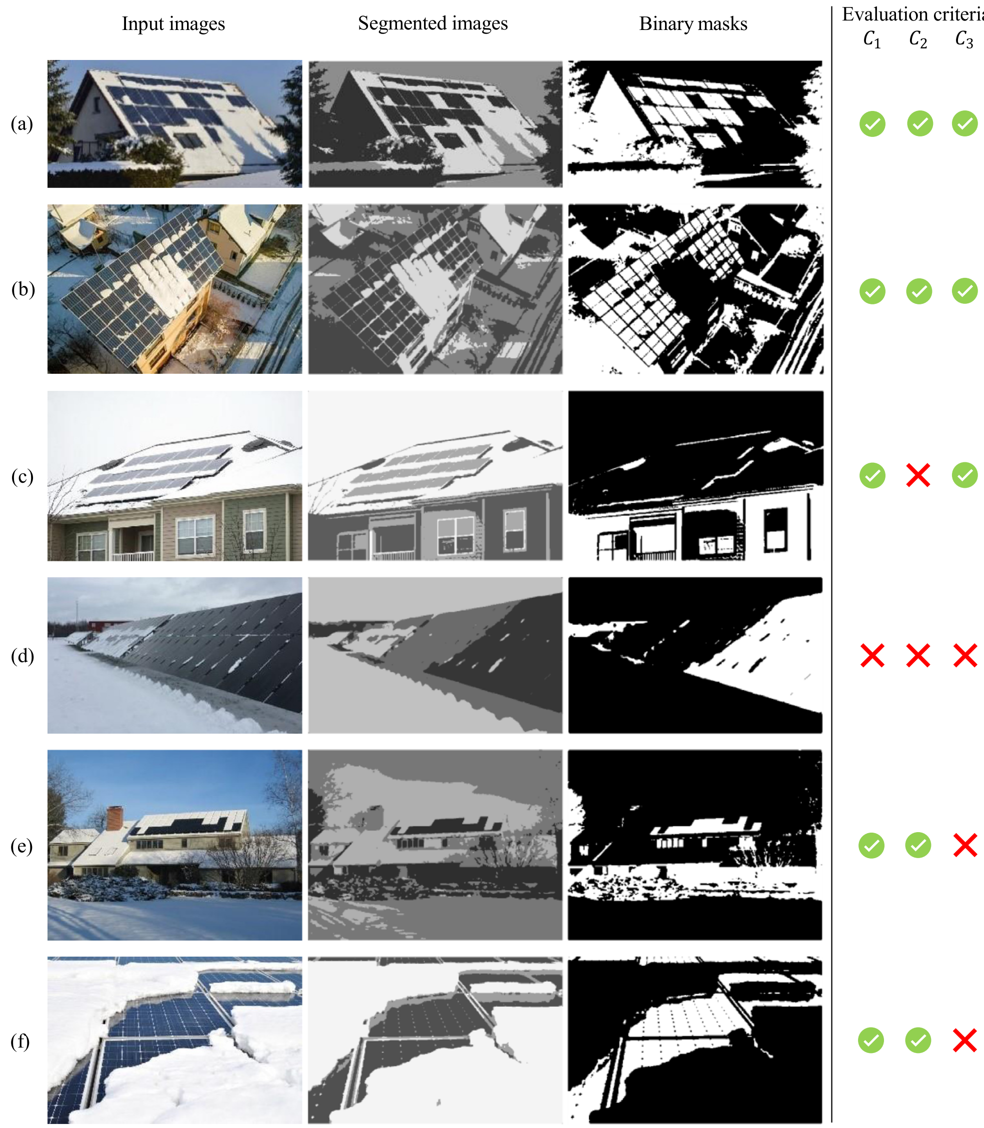


Fig. 11. Algorithm performance of six representative images. . (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

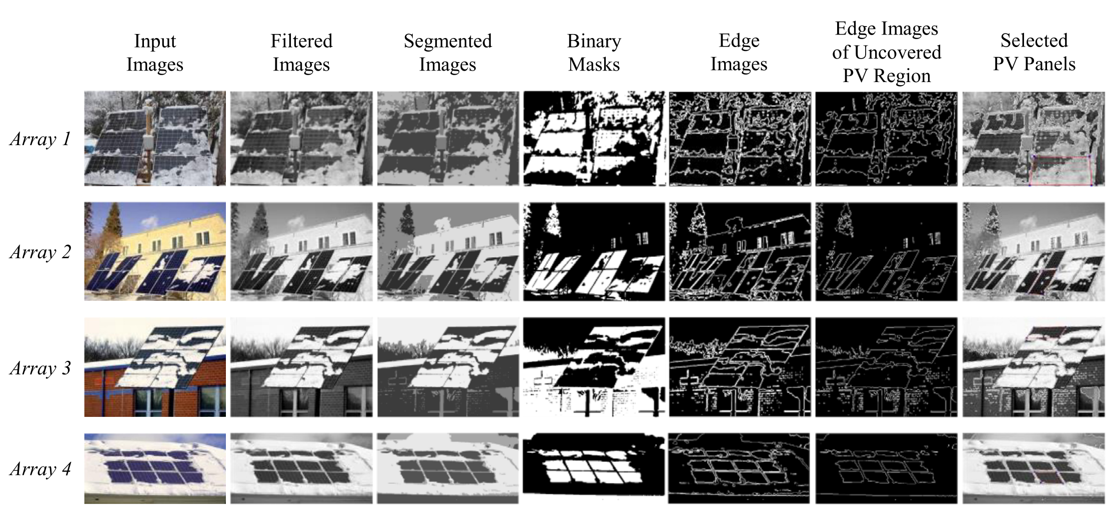


Fig. 12. Processing sequence of four images.

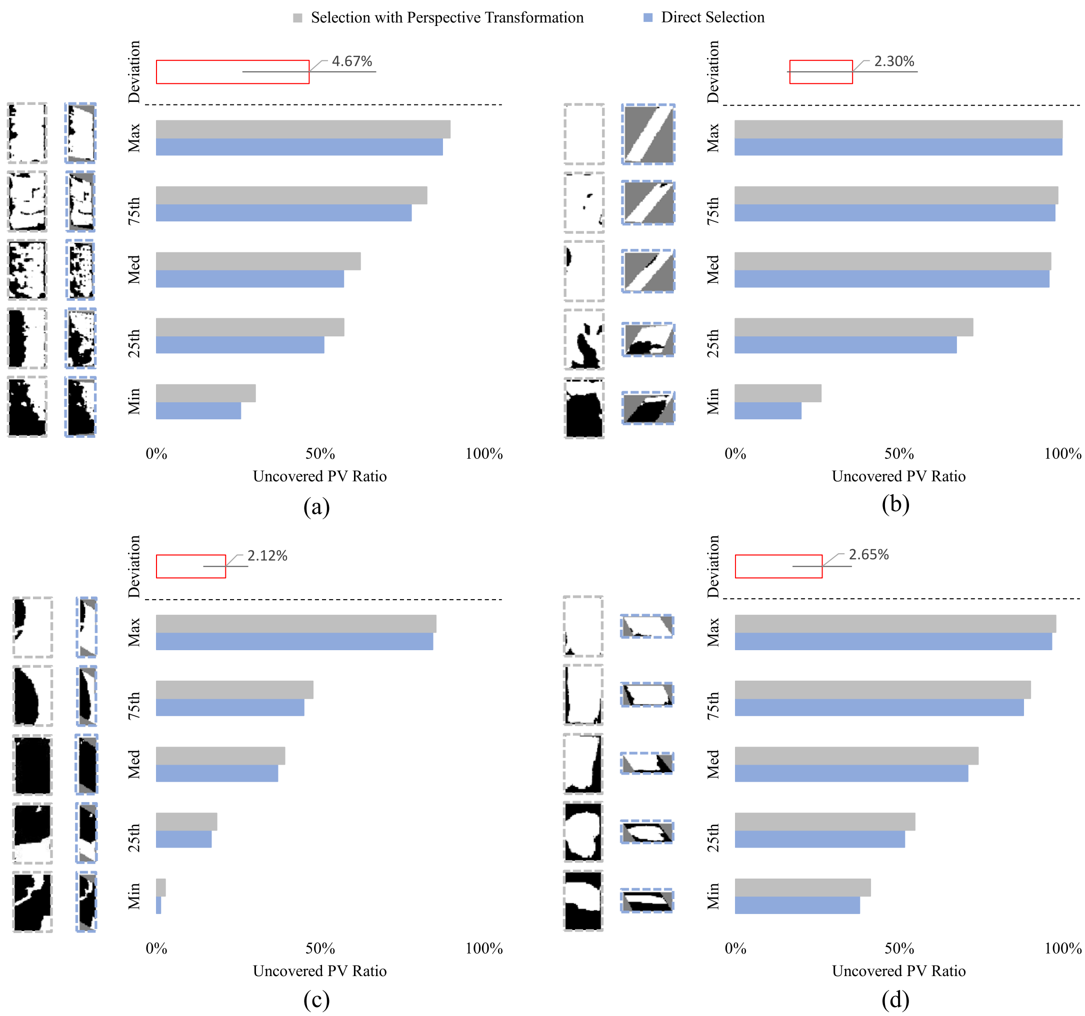


Fig. 13. Result comparison of both selection approaches (DS and PT) with processed image samples for quartiles, medians, and extremes. (a) *Array 1*, (b) *Array 2*, (c) *Array 3*, and (d) *Array 4*.

the assumption that snow layers were assumed to be consistent on an hourly basis. It is obvious that *Array 2* had the lowest average E_{loss}, which was estimated by the direct selection method and the perspective transformation method to be 0.0368 and 0.0395 kWh/panel, respectively. This was attributed to *Array 2* having the highest uncovered PV area percentage with an average value of 84.10%. Such a percentage was 36.64% for *Array 3*. Consequently, *Array 3* had the highest average E_{loss} of 0.1495 and 0.1546 kWh/panel calculated by the direct selection method and the perspective transformation method, respectively. Overall, the results provided an effective and highly accurate estimates to detect the snow cover on the PV panels with exact outputs of proportion and area. It can be further combined with a similar reconfiguration process to facilitate the enhancement of PV systems generation.

3.3. Validation with alternative methods

Fig. 15 shows the performance of the image processing algorithm for DS and PT compared with Pattern Detection with PV proposed in [22]. Six different images of PV arrays were used as the input, and the pixel ratio of the covered area in the arrays was calculated respectively. In Fig. 15(a), the deviation of the DS approach ranged from 0.09% to 3.56% of the PV array area, while the deviation of the PT approach was limited to the range of 0.72% to 1.72%. Both approaches showed a similar trend with acceptable deviations from the method in [22]. However, a relatively higher deviation was noticed in *Array 2* due to the limited detection of some frames inside the panels by [22]. This was avoided by both applied approaches, DS and PT. Fig. 15(b) shows the comparison of the shadow detection results. The generated masks of covered PV were very similar to the gray level detection results in [22], which showed the effectiveness of detection for the employed approaches.

3.4. Analysis of DL methods

Given the regression with both LMB and CNN, Pearson correlation coefficient which is generally denoted by R measures the correlation between targets and outputs. For positive correlation, R value closer to 1 shows a close relationship, while those closer to 0 show a random relationship. This can be used to estimate the error between targets and outputs, and evaluate the predictive accuracy of LMB and CNN methods. Fig. 16 shows the regression lines with R values for training, validation, test, and all for both approaches. LMB R values were 0.93, 0.55, 0.82, and 0.86,

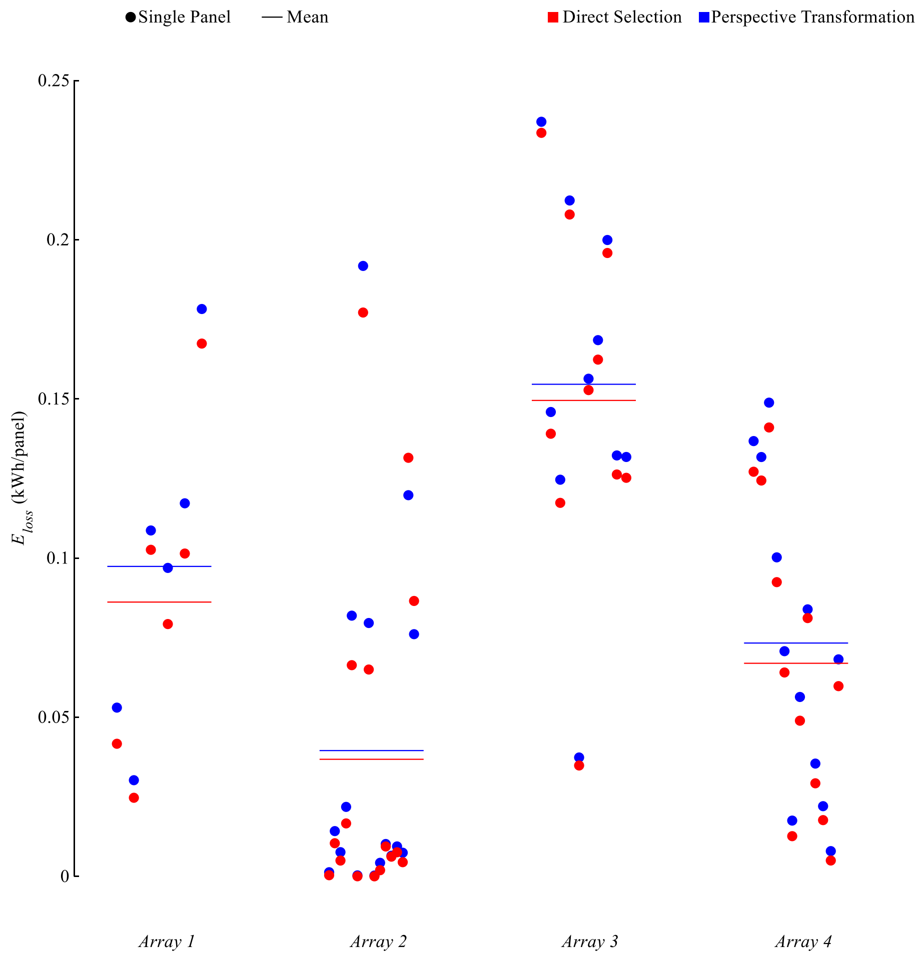


Fig. 14. Energy production loss for 44 panels in each array.

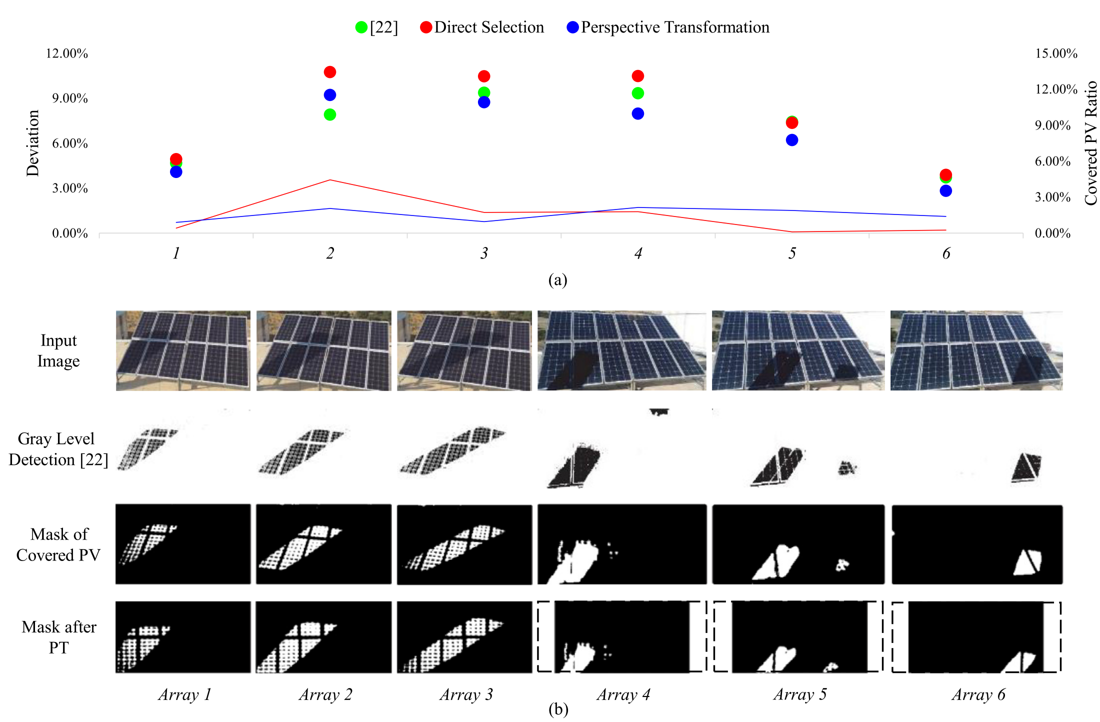


Fig. 15. Comparison between DS and PT with Pattern Detection proposed in [22]. (a) Covered PV ratio comparison with deviation. (b) Covered PV detection comparison.

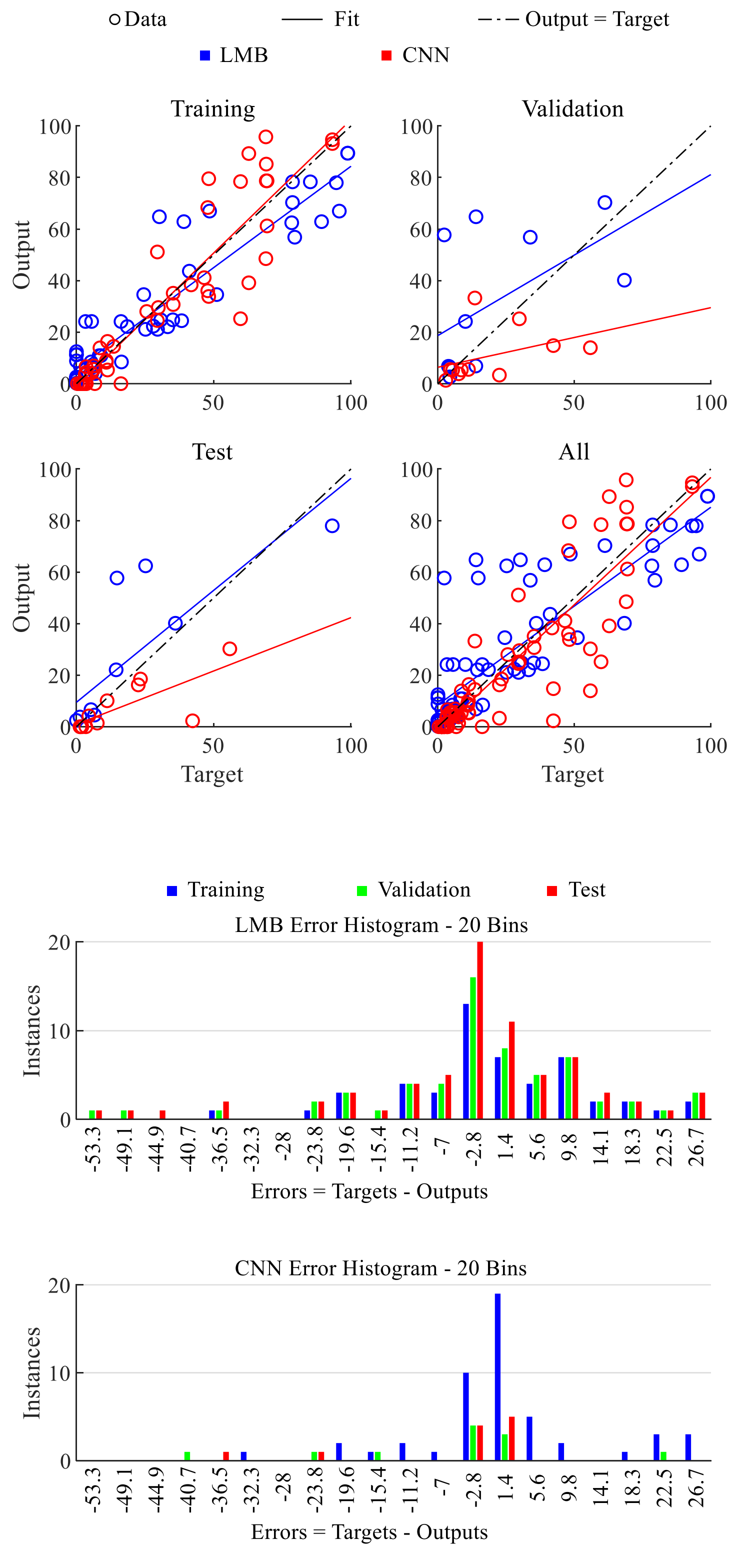


Fig. 16. Regression lines for training, validation, testing, and all sets (Top). Error histogram of the four sets (Bottom).

respectively. CNN had comparable correlations for training and all sets with R values of 0.92, and 0.94. However, validation and test sets showed weaker correlations for CNN compared to LMB. The error histogram in the same figure shows the number of instances for each error for LMB and CNN. The error values can be negative as they indicate how predicted values are differing from the targets. Neural network's reliability was investigated by monitoring the testing set. In addition to the strong predictability proved by the average R value of 0.9 of both approaches, most of the testing samples were trapped in the error range of −2.8% and

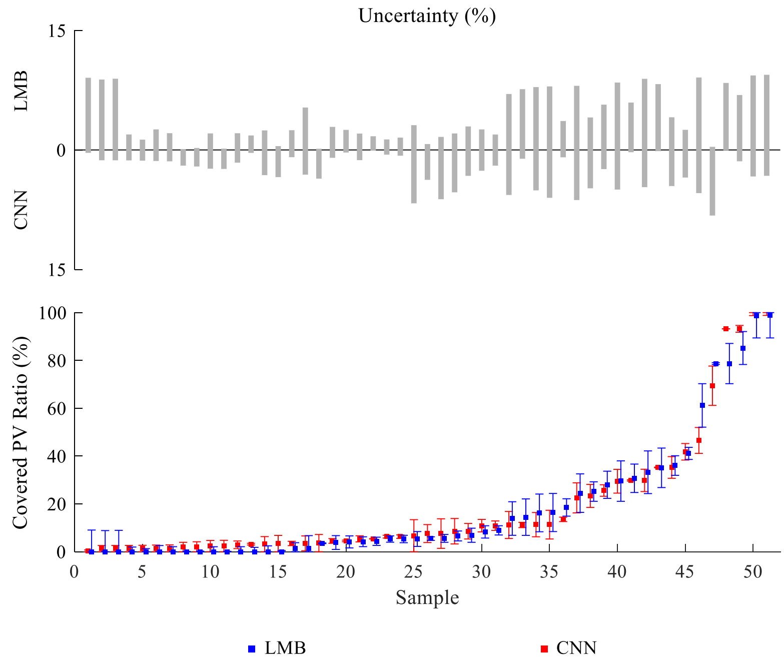


Fig. 17. Top, uncertainty values for various samples with boundaries of ±10%. Bottom, predictable range of covered PV ratios for both LMB and CNN.

1.4% for LMB and CNN. This range shows the utility of neural networks if coupled with image processing techniques that had some error values exceeding 6%. Such approach has been confirmed in [48], where deep learning compensated for image processing drawbacks such as sensitivity to the images background.

3.5. Uncertainty analysis

Uncertainty analysis was performed by estimating the deviation of the predicted PV covered area from the experimental data. Fig. 17 represents the uncertainty within the range of ±10%, where almost 70% of the examined samples were considered reliable. The uncertainty was minimal at lower covered area ratios due to negligible snow deposition. LMB uncertainty varied between 0.1% and 9.1% for covered area ratios less than 8.4%. The mean uncertainty was 2.7% over the previous range. CNN showed better performance for PVs having little snow deposition with a mean uncertainty of 1.8%. The uncertainty range for CNN over this span varied from 0.1% to 6.7%. This performance was further maintained by CNN for PVs with noticeable snow coverage portions. For snow coverage ratios of more than 9%, LMB uncertainty varied between 0.4% and 9.4% with an average value of 5.6%. Corresponding values for the CNN approach were 0.1%, 8.2%, and 3.7%, respectively. This precedence of CNN over LMB was justified by the algorithm's capabilities of detecting the important features automatically without excessive supervision, as noted by [49]. In general, the uncertainty increased at higher coverage areas with a mean value of 3.7% compared to 1.8% for limited snow deposition for both approaches. Such an overall average of 2.8% was comparable to the 3% concluded by [50] that assessed the predictability potentials of deep learning-trained models.

Conclusion

This paper presented a new method to detect snow coverage on PV panels in order to quantify the impact of accumulation on energy loss. Using image processing and recognition techniques, the study calculated the ratio distribution of deposition on the PV panels which was deemed to be influenced by multiple factors. Such factors can be recognized through identifying the electric generation deficiency using real-time images. The characteristics of the images applicable to the proposed algorithm were analyzed with corresponding reasons. The experiments were conducted on 44 different PV panels in four arrays that housed various ambient conditions for PV installations. The direct selection and perspective transformation techniques were compared to previous models where their average deviation was 1.17% and 1.25%, respectively. The performance of both techniques was assessed and compared, with an acceptable deviation that ranged from 0.1% to 7.35% of one panel area, yielding the effectiveness and high accuracy of both approaches despite the slight distortion in direct selection method. Energy production loss was computed to show its negative correlation with uncovered PV ratio. Further experiments also verified the accuracy of deep learning techniques in PV performance prediction, demonstrating the potential of combining deep learning with image processing.

This paper breaks the limits of specific application scenarios for most current studies [27,28,51], and proposes an algorithm that is universally applicable to various PV conditions. It is also the first attempt to employ two different PV selection approaches while evaluating the performance of deep learning. With the work in this study, effective PV area was identified and energy production loss caused by partial or full snow coverage on PV panels was assessed. Combined with electrical circuit reconfiguration strategies, it would facilitate PV systems to achieve maximum power generation efficiency, which is of vital importance to regions under cold weather with snow deposition. Meanwhile, the improvement of energy efficiency due to the remote control over PV systems facilitates solar energy to take the place of fossil energy. This study can extend in the future to involve the integration of deep learning with cognitive technology to help identify and categorize unstructured data in the PV arrays into specific classifications to detect layouts patterns with increased efficiency and accuracy. Future work should also consider applying more data samples to address the study limitation in relation to data shortage. Additional applications of statistical analysis can lead to advanced inferences that reasonably extend the findings from the existing image data sample.

CRediT authorship contribution statement

Xinyi Zhang: Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization. **Mohamad T. Araji:** Conceptualization, Software, Formal analysis, Resources, Writing – review & editing, Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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